

Systemic Intelligence

SysQ™



*Increasing Our Capacity to
**Build Understanding and
Improve Performance***



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consulting

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“If a factory is torn down but the rationality which produced it is left standing, then that rationality will simply produce another factory. If a revolution destroys a government, but the systematic patterns of thought that produced that government are left intact, then those patterns will repeat themselves... There’s so much talk about the system. And so little understanding.”

—Robert Pirsig

Purpose

This document is designed for anyone – individual, organization, coalition, community, or nation – who is considering applying systems thinking or system dynamics to solve complex problems, make better decisions, develop high impact strategies, and build adaptive leadership capacity. Perhaps you are exploring the possibility of hiring a practitioner – a guide for this journey

By building our Systemic Intelligence (SysQ™) – the capacity to think *and* act in accordance with systems thinking – we can:

- **Solve** the most vexing, **complex**, and seemingly intractable **problems** we face
- **Engage** others in a process of **transformational learning**
- **Create solutions** that will be understood and **implemented by all** stakeholders

This document will help you understand how to learn and apply the larger framework of Systemic Intelligence (SysQ) – and why **SysQ** is a more useful term for **practical systems thinking**. The document is organized in the following sections:

- **SysQ Overview** – A quick definition of Systemic Intelligence
- **The SysQ Value Proposition** – Why SysQ is an essential capacity for today's complex world
- **SysQ vs. Systems Thinking** – Why Systemic Intelligence should replace the term systems thinking
- **SysQ Goal = Leverage** – The purpose of SysQ is to find leverage, so what's leverage?
- **SysQ Components** – What are the three capacities required to apply SysQ
- **SysQ Process Details** – Tips for best application of SysQ
- **SysQ Facilitator Role** – What to expect when working with a SysQ practitioner

If you read only one section of the document, **read** the section titled: **SysQ Facilitator Role**. Clients have baked-in **biases** that run counter to the **SysQ** approach, generating **tension** between client and consultant. The SysQ Facilitator Role section explains several of the tension points between typical problem solving and the **type of problem solving** SysQ requires.

How to Learn and Apply SysQ

Working with a SysQ practitioner can help improve the following: **diagnosis**, **intervention**, and **capacity building**.

- **Diagnosis:** Accurately **diagnose** and **frame** problems in a more actionable way
 - **Intervention:** Design **innovative solutions** for addressing problems in the most **efficient** and **effective** way.
 - **Capacity building:** Focus on **skill transfer** – not only helping solve a pressing problem, but also helping build capacity for **solving future problems**.
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The Power of Systemic Intelligence

- * A major aerospace company's internal software development team faced cost overruns and anemic customer adoption. Applying a more holistic planning process led to high customer ratings and \$10 million savings.
- * A top metro region struggled to attract skilled workers. Importing senior workers exacerbated the problem. Applying a systems-based process, a coalition developed a unique strategy – it not only solved the shortage, it became one of the US's top five most desirable cities.
- * For decades, the World Bank supported a developing nation with persistent malnutrition. A multi-stakeholder team applied collaborative learning based on systems analysis. Malnutrition halved within 10 years.
- * A father of four daughters struggled to support his youngest's fear of failure. Public praise created jealousy with a second daughter – who teased the youngest – undermining his strategy. His 50,000-foot analysis identified the problem...and he developed a private approach. He even enrolled the teaser.

SysQ Overview

SysQ, or Systemic Intelligence, is the capacity to **understand complex systems** and **find high-leverage solutions**. It involves a **mindset** focused on understanding system structure and higher-order **thinking skills** like DYNAMIC, FOREST, and OPERATIONAL Thinking.¹ SysQ helps identify **interventions** that optimize resource investment *and* improve ecosystem performance. SysQ **improves our understanding** of complex systems by developing useful, high-impact **mental models**.

Mental models are like **mental maps** we use to make sense of the world – to help us navigate the vicissitudes of life. Mental models are theories or sets of **assumptions** about how the world works: based on what we know, what we've experienced, and how we think. **They affect how we see things, make decisions, and act.** And they are **imperfect** – they can leave out important details or be biased – so we need to constantly **test** and **improve** them.

The SysQ Value Proposition — Why It's Needed

In any sector of life, **we** must make decisions, develop plans, and **implement strategies**. Many succeed. But...some will fail...fall short. Some will end **catastrophically** – like reorganizing in response the Challenger disaster didn't prevent the Columbia's explosion on re-entry; or the US COVID strategy led to at least 200,000 unnecessary deaths.²

The **world** is becoming **increasingly complex** and we can feel overwhelmed by the challenges coming at us.

- Personal health and economic concerns
- Communities struggling with racism, homelessness, and poverty
- Global unrest with spreading authoritarianism and conflict
- Climate change is here and accelerating its impact

Solving these challenges can feel **overwhelming** to the point of reaching exasperation – acquiescing to an inevitability. However...

...in each of the four successful examples described on the previous page, **teams and individuals created innovative, sustainable solutions by avoiding common pitfalls in thinking**, and instead applying a powerful capacity: **Systemic Intelligence**, or SysQ. Building and applying **SysQ** can help solve even the most overwhelming of challenges.



"Most people do not fully appreciate the systemic nature of their problems. We think and act parochially. The cultures of our respective groups and the respective roles we play in these groups often cause us to view problems through the narrow and myopic lens of immediate self- or group interest."

—Dean Williams

SysQ Vs. Systems Thinking

The reason we each feel overwhelmed – burned out – is because we’re dealing with what Russell Ackoff refers to as a **mess** – a **system of systems**.³ It’s appropriate to call many situations messes!

You’re not just working on a system. You’re working on an **interconnected set of systems**.



“Managers are not confronted with problems that are independent of each other, but with dynamic situations that consist of complex systems of changing problems that interact with each other. I call such situations messes... Managers do not solve problems, they manage messes.”

—Russell Ackoff

If you’re trying to improve your organization’s bottom line – perhaps while focusing on aggressive sustainability goals – you’re not just working on the manufacturing system or the service delivery system. You also have to work with human resources, supply chains, external partners, competition, and uncertainty in your customer base. **Each is a system**; improving performance of the whole requires understanding how those system of systems work together.

Several states are currently in the process of improving behavioral health outcomes. Of course, this means working with behavioral health systems. But...it also **requires engaging** and **improving** the performance of **systems** in education, public safety, the judiciary, general medicine and even how advocacy groups and families contribute to ecosystem performance.

Although the term mess is appropriate, I prefer using the term **ecosystem transformation** to describe the challenge of improving a system of systems. Many of my colleagues still refer to ecosystem change work as systems thinking. This term has been diluted over years to the point where many clients **think** they **apply systems thinking** simply because they *work on a system* like claims processing. That’s not it.

Systemic Intelligence – **SysQ** – is the capacity to **understand ecosystems** we wish to improve with the level of insight needed to fundamentally **improve** them. Like emotional intelligence, we all have some level of SysQ; but the world is becoming increasingly complex – where more and more of our crucial problems are what Ron Heifetz calls **adaptive challenges**.⁴ Our capacity to address adaptive challenges is woefully inadequate relative to the need.

SysQ Goal = Leverage

The goal of SysQ is to generate **high impact** when solving problems and creating strategy. The SysQ Goal = Leverage. To find and implement **leverage** is to:

Fundamentally improve ecosystem performance, while optimizing the investment of limited resources – where possible – and simultaneously anticipating and minimizing unintended consequences

SysQ Components

There are three interconnected components of the SysQ Framework:

- MINDSET
- THINKING SKILLS
- PROCESS

Mindset

The MINDSET is summarized by the core tenet of SysQ

To Find Leverage You Must First Understand the Structure Driving Behavior

Let's clarify what that tenet means...

EXAMPLE — System 1

I'm going to administer an input to this system. Answer the following question with the first idea coming to mind.

What's the cause of this behavior?



Click on the image to open a YouTube video in your browser.

Before going to next page please answer: **What's the cause of this behavior?**

If you're like most people, you answered: **gravity**. Some will answer **removing the hand**. Here's another system. I'll administer the same input.



Click on the image to open a YouTube video in your browser.

Huh? Different behavior. Thankfully, because my feet are still dry!

So gravity and removing the hand are necessary but insufficient to create the oscillating behavior.

Answer again. What's the cause of this behavior? Review the previous video if needed.

Most people will now say it's the slinky...**it's the system**. A slinky is an oscillation waiting to happen – which happens when I remove my hand.

One important attribute of leverage is the ability to fundamentally improve ecosystem performance. Imagine that you want to significantly improve the oscillations – to dramatically dampen them – perhaps they are creating chaos in your organization.

If your mental model is that gravity is causing the oscillation, how much leverage do you have? None. You can't change gravity. I imagine you can identify times in your life and work where it feels like fighting gravity – something refuses to improve no matter how hard you try.

If your mental model is that removing the hand is causing the oscillation, how much leverage do you have? Before you say some, let's also say you don't own the hand. Again you have no leverage. You can either become an Eeyore and proclaim nothing will change. Or you'll be a victim and blame others. Either way, you're stuck with the problem.

However, if your mental model is that the **system generates the behavior**, you can **modify** the **system** so that when the hand is removed by someone else, even though gravity exists, the **behavior** is fundamentally **improved**. You've found **leverage**.



Click on the image to open a YouTube video in your browser.

EXAMPLE — System 2

Watch this cotton candy eating contest.⁵ Whose mental model is the high-leverage one? Although I personally relate to how the guy on the left feels, let's focus on the winner. He had a mental model that deeply understood the structure of cotton candy. He was able to modify that structure to significantly accelerate the time to eat the cotton candy.

We often respond to problems using the same old thinking – applying **outdated** and **inaccurate mental models** – without once questioning: Do I know the structure of this ecosystem that's responsible for this problem?

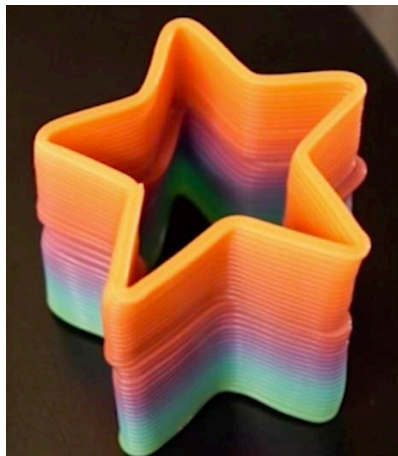
If we don't develop **sufficient systemic insight** into how the **ecosystem works** we will be unable to achieve the performance goals we set. We will continue to be buffeted by the "slings and arrows of outrageous fortune".



Click on the image to open a YouTube video in your browser.

The Focus is *Only on the Structure* Generating Performance

Disciplined application of SysQ means we **focus only on the structure generating performance**. People with high SysQ first filter out the essential from the mess – from the noisy barrage of data – to **cogently establish** what **performance** they wish **to transform**, to see unfold in the future. Then they assemble a mental model of the **structure** that can **generate** that **performance**.



Having high SysQ means avoiding wasted time in trying to understand EVERYTHING about the ecosystem. Remember the slinky: all I need to know to improve performance is to **focus on the structure**. If I tried to understand the whole system I'd needlessly spend time analyzing the **colors** or the **shape**. Similarly, with the cotton candy system, although perhaps of great importance to the eater, the flavors or colors don't contribute one iota to how the structure behaves. Yet many of my clients want to spend time analyzing interesting aspects of their ecosystem – even though those aspects are irrelevant. Studying them consumes precious time and resources...often leading to the dreaded **analysis paralysis**.

"The important thing in science is not so much to obtain new facts as to discover new ways of thinking about them."

–Sir William Bragg

We Often Focus on the Low Leverage

Organizations often focus on **low leverage** interventions, investments, and policies. They will adjust price up or down. They'll **tweak** a complex budget a few percentage points one way or another. They'll **replace** one person in a **leadership position** with a different one – without changing how the rest of the organization around the individual works with them.

These low leverage interventions **cannot fundamentally change behavior** – changing the colors on the slinky would change its appearance, but it will still oscillate.

Our low leverage strategies **overlook** more **transformational interventions**: radically modifying rewards and incentives; changing the type of data generated – including who does...or doesn't...have that information; and shifting fundamental paradigms – like competition vs cooperation.

SysQ helps **identify** those **higher leverage** interventions often **overlooked** when developing strategy. Remember, the basic tenet of SysQ – the underlying mindset – is:

To Find Leverage You Must First Understand the Structure Driving Behavior

Thinking Skills

We need **higher order thinking** to follow the mindset or core **tenet** of SysQ – to **understand structure driving behavior**. This requires

building rarely used **systemic thinking skills**.⁶ We must shift up from the traditional thinking we apply to problem solving into this higher order thinking. Some types of shifts in thinking include shifting from...

“In history, good intentions do not always make good consequences”

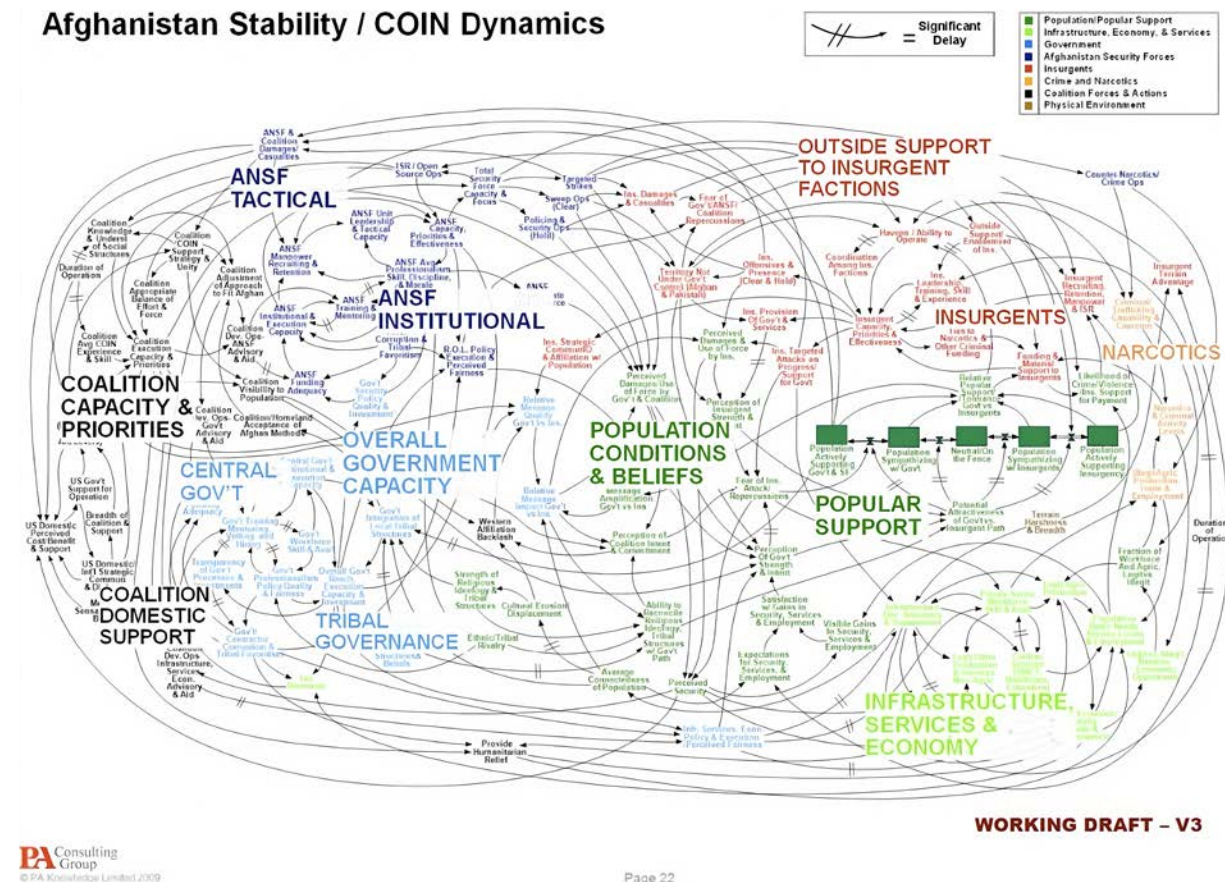
–Alija Izetbegovic

Typical Thinking	SysQ Thinking Skill
EVENT Thinking We focus on the latest or most newsworthy event.	DYNAMIC Thinking We shift to seeing how that event is part of a longer term trend. This helps us more clearly see what structural forces might be driving that pattern.
TREE Thinking We focus individually on each silo in the organization, each department in the division, each sector in society, each country around the world – the equivalent of tree-by-tree thinking.	FOREST Thinking We try to understand how each silo interacts within the organization and even outside it. This interconnected web of departments, divisions, offices, silos, and sectors – the forest – is responsible for larger system behavior.
CORRELATIONAL Thinking We use regression and correlational analysis to propose causality. Or use anecdotal evidence of when I see this happen something else happens – without a clear and rigorous causal theory.	OPERATIONAL Thinking A former colleague read an esteemed academic paper on US dairy policy that was prescribing potential strategies to increase production. The analysis included economic indicators like interest rates, feed prices, farming expenses – but despite having high statistical confidence, the analysis left out the one system component essential for production. Cows. Only a model that includes cows can be operational. Only operational models create clarity on levers. Yet we often create strategies based on non-operational thinking as useless as a dairy model without cows.
LINEAR Thinking We assume working more hours will complete more work, hiring more people will impact client satisfaction, hiring a consultant will improve our capacity, and an endless stream if IF THIS THEN THAT.	FEEDBACK LOOP Thinking The world works with feedback loops. For example, there’s a feedback loop where more hours worked builds burnout, less completion, more hours, and more burnout. A nasty feedback loop begins to dominate as a vicious cycle. Feedback loops drive the complex dynamic behaviors experienced in the real world and must be included in our strategies and decisions.
MEASUREMENT Thinking A commonly held belief is to make decisions only using data that we can measure precisely and represent numerically in our models. Production can be included. Burnout can’t.	QUANTITATIVE Thinking Anyone think Burnout doesn’t determine production? Unless you do believe it has no impact, then you know any model excluding it to be wrong. Even if unmeasured, it can still be quantified – even if anecdotally – ranging from 0-100. 0 means everyone is super relaxed; 100 means they are immobilized from burnout.

EXAMPLE — SysQ THINKING SKILLS

Let's look at how to apply the thinking skills described here in the context of the US Afghanistan counterinsurgency strategy. We'll use an analysis by Barry Richmond – originator of many of the SysQ thinking skills – and further described by my colleague, Steve Peterson.⁷

When asked to develop a systems analysis of the counterinsurgency strategy, one consultancy drew this map.⁸

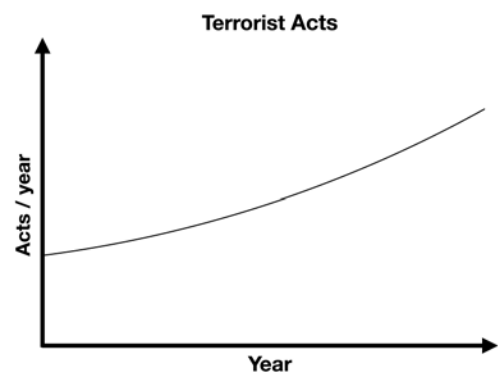


Although it is a valiant effort at mapping the system, it doesn't conform to the principles of focusing only on the systemic structure responsible for the behavior. Instead it's the equivalent of mapping every color in the slinky previously shown and including the shape and other attributes that don't contribute to the oscillations.

Applying the thinking skills would work as follows.

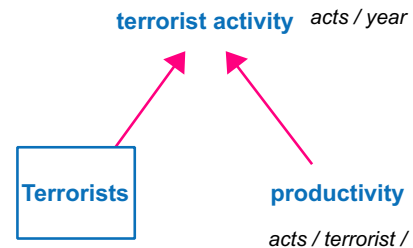
Instead of looking at the latest terrorist event as something to respond to in isolation – where a strategy would be to hunt down those responsible and bring to justice – **DYNAMIC Thinking** would suggest it's important to look at **terrorist acts** as a **trend over time**.

Further, **FOREST Thinking** would ensure that the activity of terrorism wasn't just localized to a region



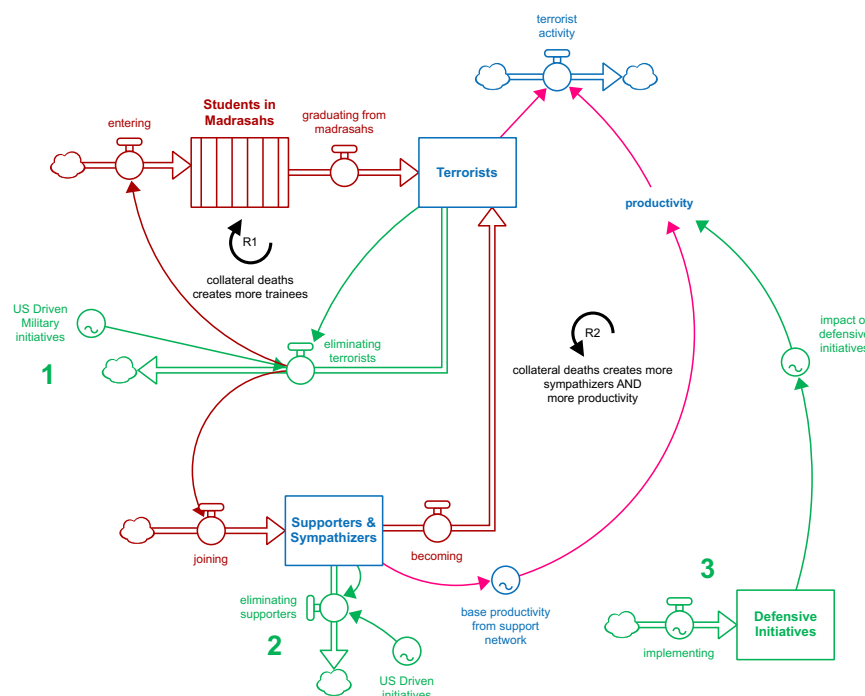
within Afghanistan. Instead it would **look** at such activity over the **breadth of the country**... and perhaps the **network** extends even **outside the borders**.

Having used SysQ to frame up the issues to be able to find leverage, we can apply OPERATIONAL Thinking. The number of terrorist acts occurring over a year is generated by **two variables**: the **number of terrorists** and their **productivity** (terrorist acts / terrorist / year). By making it operational, it becomes clear there are only **two levers** (places) to apply an intervention to reduce terrorist activity. **Reduce** the number of **terrorists** and or **reduce** their **productivity** – their terrorist acts/terrorist/year. That's it. All solutions will be designed to work on **one or both** of these **levers**.



LINEAR Thinking would suggest that an intervention that kills terrorists would work to reduce terrorist activity. However, by applying **FEEDBACK LOOP Thinking** we can see it will likely kill people who are not terrorists; this will **motivate more** potential **terrorists** to enroll in terrorist organizations and eventually be trained to commit acts. This would create a nasty **reinforcing feedback loop** (R1) known as a **vicious cycle**.

Finally, although in our simplistic view of the world, it would be awesome for our actions to only impact the intended target, we just learned with FEEDBACK LOOP Thinking that's often not the case. In fact, by applying **RIPPLE EFFECT Thinking** we can see that not only would killing terrorists likely lead to more terrorists signing up. Killing innocent civilians can also lead to **more sympathizers**, who can increase terrorist **productivity**. They might provide financial resources, hide or house terrorists, or share intelligence that helps them.



In our organizations, we assume that lowering **price** can increase sales. But **FEEDBACK LOOP Thinking** can help us see that sometimes if **sales** exceeds capacity, **customer satisfaction** falls, and **sales** then **decline**. That's a counteracting or balancing loop.

In our communities, we believe that we can address crime only through effective policing. But there are other **aspects** of the **community** that **contribute: housing,**

unemployment, access to **substance abuse support**, and even community belonging and **social capital**. This FOREST Thinking approach is needed to have effective solutions.

There are many thinking skills that are part of the larger cognitive capacity I refer to as Systemic Intelligence, **thinking skills** that make up **SysQ**. These thinking skills are **useful** in **all aspects of life**. But in order to make better strategies –in order to find leverage – we must apply them with a **process** and set of **tools**.

SysQ Process and Toolkit



Finally, even if we adopt the mindset and build the skills of SysQ, we still need a **process to apply** them. Fortunately, there's an easy to follow **three step process** for **finding leverage**. These three nonlinear, iterative steps – or phases in the “finding leverage” process – are:

- A. Sensemaking**
- B. Solving**
- C. Implementing & Learning**

“The world is a complex, interconnected, finite, ecological-social-psychological-economic system. We treat it as if it were not, as if it were divisible, separable, simple, and infinite. Our persistent, intractable global problems arise directly from this mismatch.”

–Donella Meadows

A. SENSEMAKING

We must first understand the underlying forces creating the problem or driving the performance we want to change in order to improve it. A **common pitfall** is that **we** often **assume we know** a problem's **cause**...we **move** straight **into** the **solution phase**.



However, there's a much more **insidious** root **cause** to **ineffective strategy**. We often **choose** the **wrong issues** to work on. That's right...we're working on the wrong problems! We've **poorly framed** the **problem**.

Assume I want to improve my child's grades. I may assume what I'm trying to do is improve their focus and persistence on studying. So I'll incentivize them with money. This may work in the short term. But the likely real problem to fix is their underlying intrinsic motivation. And the cause of that might be poor self worth or efficacy. If that's the problem, then the reward solution was a mismatch with the right problem.

This is the equivalent of **building a great ladder**... and then leaning it **against** the **wrong wall**! In order to make sense and build a great strategy, we must first **choose** the **right performance issue** to improve – we must choose the right wall.

Even if we spend the time to frame the right issues, we still **often** just **jump to solution mode**. We just begin building our favorite ladder.

In the grades example, we might move into reactive mode and resort to words of encouragement. You're smart. You can do it. Only to find that our child rejects those words... even arguing with us. We might then get frustrated. The wall might include a fixed or stuck mindset: I'm bad at math.

By **identifying** the fixed mindset is the **type of wall** we're climbing we **can** then **build** the **right ladder**. Instead of using the labels of "you're good at math" – which my child doesn't believe – focus on learning. "Yes, you might not be as good at math as you'd like, but you can learn. In fact, even though you still got a C on the latest test, it's several points higher than the last." Internalizing the "I'm learning" message – although slow in progress – will ultimately create a lifelong learner that accepts and enters challenges in effective ways.

Successful **sensemaking** requires first **framing** the **right performance issues** to improve **and then building a systemic picture** of the **structural forces** driving performance. We are **averse** to – or often don't diligently apply – **either step**. We adhere to "Don't just sit there, do something". But, the right frame of the issue **and** a rigorous operational picture of structure must come first; next you can generate higher leverage solutions...and then act.

"If you can define the problem differently than everybody else in the industry, you can generate alternatives that others aren't thinking about."

–Roger Martin

B. SOLVING

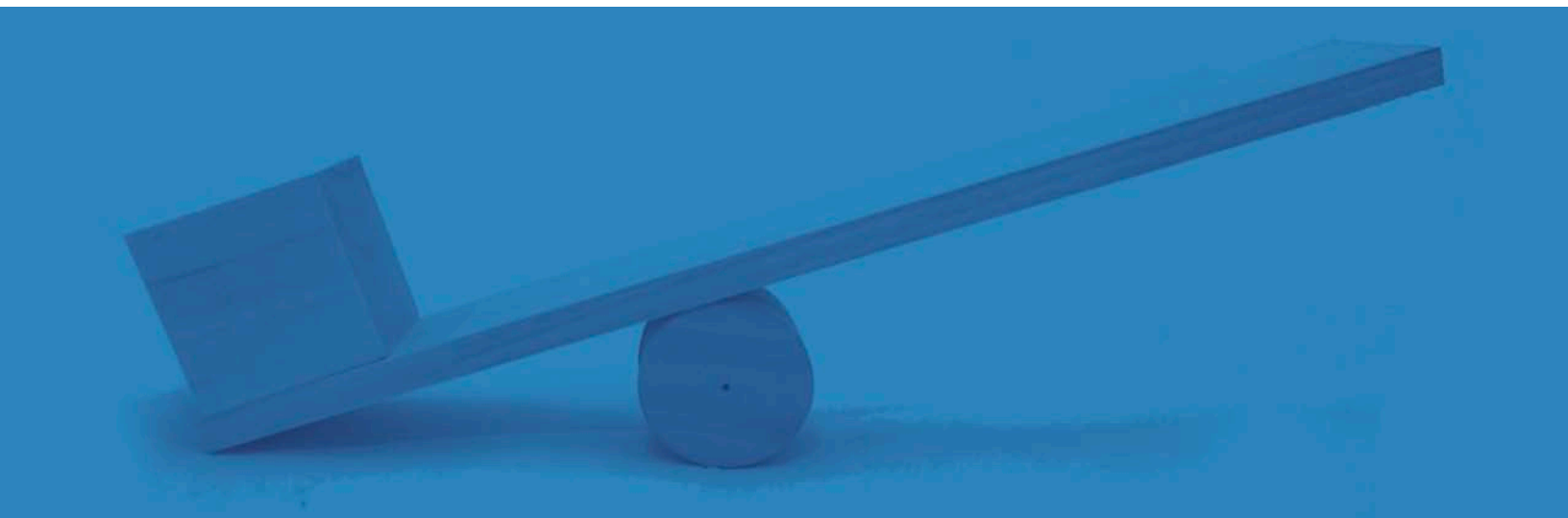
Assume we have successfully chosen the right performance issue to improve – we’ve framed the right problem. And then we have accurately modeled the structural drivers of performance. It’s then, and only then, that we can look for leverage.



If applying high SysQ, we will look for places to **intervene** that can **fundamentally improve ecosystem performance**. In addition, where possible we pursue **optimizing** the investment of **limited resources** and while looking out for **potential unintended consequences**.

When startup organizations enter the market, there’s a reflexive, never questioned strategy that it’s essential to enter with a low price. As previously discussed, low price can certainly increase sales. It can also create **demand** that **exceeds** current **capacity** to meet that demand. This will create delivery delays, product defects, and **poor customer satisfaction** – thus decreasing future sales. Often, a high-leverage strategy is keeping price high enough to meet demand from only early adopters who want new features and services, yet less concerned with ease of use and slick implementation. The organization can use profits to first improve capacity and then slowly lower price so that demand and capacity remain balanced.

Most of the time we **choose** from **solutions** we’ve **already proposed** or even **tried**. We find it hard to be creative and think outside the box. Instead, we get **more of the same**. Slightly more or less customer satisfaction. A small decrease in unemployment...or a slight increase. Less carbon emissions by a small percentage, but not close to our espoused goals.



“The framework, tools, and language of systems thinking [Systemic Intelligence] should be accessible to all. Anyone can do this at some level, and everyone should try!”

–Barry Richmond

SysQ can help improve our ability to generate solutions that are:

- Innovative and Overlooked
- Counterintuitive
- High-leverage

Innovative and Overlooked

Focusing on interrupting the madrassas pipeline training new terrorists is an overlooked strategy – it's far more innovative and outside the box than typical anti-terrorism strategies.

When a metro region chose to create a more robust arts and culture scene to address workforce shortages...that was an innovative economic strategy.

Another innovative strategy? Recategorizing internal customers into On the Fence, Advocates, and Haters and applying a strategy to reduce creating early Haters – instead of the normal focus on building early Advocates – saved the aerospace company millions.

We can generate all of the above **innovative**, outside the box **strategies** precisely **because** we've framed up the **right issues** and developed the **right map** / model of **structural drivers**. We **see** more **broadly** and we understand more **operationally**.

Counterintuitive

Often we choose to work on the right levers, but rather than receiving a pat on the back for a positive impact, we experience the ecosystem responding in counterintuitive ways. Sales leadership is shocked when they pull the price lever down to lower price only to find it leads to lower sales in the long run.

Building areas of solely low income housing can generate neighborhoods and communities that can become poverty pockets. It is counterintuitive to remove low income housing and replacing with mixed use and mixed income – with ranges of low to high income housing. The amount of solely low income housing is a lever. Our emotional reactions to poverty often leads us to **pull the right lever – but the wrong direction**.⁹

High-leverage

How often are you in planning meetings where everyone is adjusting last year's strategy or budget a little bit? Slightly more resources here; slightly less emphasis there. This tweaking strategy might be best called the Buckshot Strategy Planning Process. We **assume** that we need a little bit of **scarce resources applied** to **everything** we deem **important**, so we discharge in a **buckshot approach**.

Suburbanites are shocked when building new roads and adding lane miles – in an effort to reduce traffic congestion – ultimately feeds back around to greater congestion a few years in the future. Reducing congestion creates incentives for more development, which then increases congestion. Rather than trying to eliminate congestion, it's better to embrace road congestion and use that pressure to stimulate jobs closer to where people live or public transportation.

One major insurance provider was challenged to maintain sufficient claims processing staff. New hires quickly burned out dealing with understandably stressed out customers wanting

their claims awarded – and fast. The **typical strategy** is to focus on the **number of people**; we **need more** of them! So the company would **hire more**, only to find **burnout** and **turnover** continue **accelerating**. They found a **high-leverage** strategy. Develop the **emotional intelligence (EQ)** of the department, especially its senior leaders. This allowed the staff to engage with customers using **greater empathy** and generating **less burnout**. Further, senior leaders could help support their staff by **focusing** on their **well-being** – which **reduced turnover** more.¹⁰

Our strategies often focus on multiple physical aspects of the ecosystem. Increasing inventory and buffers, developing new facilities and infrastructure, and hiring more staff. It is **often higher leverage** to **focus** on the **less physical aspects**: building **skills**, changing **decision rules**, reducing **burnout**, improving **information** flows, increasing **incentives** or **penalties**, and ultimately transforming the **mindsets** we use to make all decisions.

These higher leverage interventions – although hard to find in typical planning frameworks – become illuminated when applying the skills of SysQ.

C. IMPLEMENTING & LEARNING



We move into implementation once we have confidence we've framed up the right problems to address, understand the structural drivers behind them, and identify the high-leverage solutions. But the more difficult the challenge, the more **this phase isn't the end of strategy** or problem-solving. It is **often the beginning**.

The SysQ process is really a simplification of the scientific method where all analysis prior to implementation is building and testing a hypothesis – a potential strategy – and not a truth. One defining attribute of the SysQ process should be to **move to the implementation** phase **as soon as**, but no sooner, than **possible**. Possible means spending sufficient time sensemaking prior to solving.

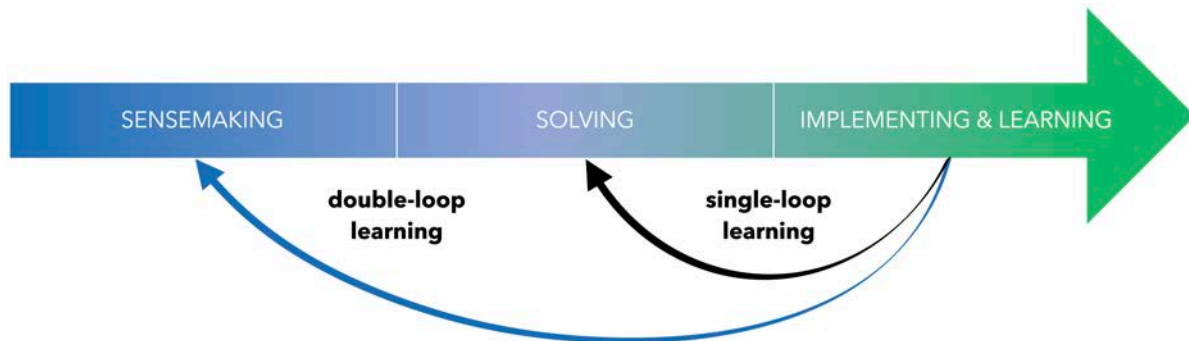
"If you want to truly understand something, try to change it."

–Kurt Lewin

We start with framing up a performance issue to understand and then build a model of that performance issue. We then generate a hypothesis for how to improve that issue, which we often call decisions, plans, or strategies. However, even though we've tested using SysQ skills and tools, we need to continue building confidence in the hypothesis. And that necessary confidence is best built as we implement the strategy.

Kurt Lewin said that **"to understand something, try to change it"**. As we learn from implementation – **see what works** and what doesn't – we can determine if we have the wrong strategy, if it needs to be **modified slightly** or **changed dramatically**. A higher leverage **learning** occurs when we understand we've made sense incorrectly. Perhaps we don't fully understand the causal drivers. Or perhaps we've chosen the wrong performance issues – we're building ladders for the wrong wall.

Reviewing and **revising** our **sensemaking** is sometimes labeled: **double-loop learning**. Single-loop learning occurs when we think we have made perfect sense of the world and only change some of our tactics to solve problems. Double-loop learning occurs when we **hold** our **theories** of the world **as a hypothesis** and are willing to **question** our **assumptions**; we are willing to **change** our **thinking before we change our actions**.



The goal of the Implementation and Learning phase of the iterative learning process should be to gather useful information as rapidly as possible to confirm or disconfirm the strategy. Find **leading indicators**, monitor them, **analyze** the information, decide whether you **have** the **wrong strategy** – where we can try adjusting strategy using the same thinking we developed it with. Or we may realize we have the **wrong understanding of the ecosystem**, or the **wrong performance issues** to improve – we may double-loop. If we need to modify our assumptions we do so, and then move through the process again.

TOOLKIT

When first delving into the concepts of systems thinking, **most focus on a few** of the **tools**: system archetypes and causal loop diagrams. The reason for putting **tools last** in this description of Systemic Intelligence is that the **thinking comes first** – **tools** are only needed in so far as they **support** the ability to apply the **SysQ Thinking Skills**.

Some of the SysQ tools I use with clients – that are applied at different phases of the process – include:

- The SysQ Questions
- Behavior Over Time Graphs
- Multi-solving Tools – e.g. FLOWER
- Systems Archetypes
- Causal Loop Diagrams (CLDs)
- Stock and Flow Main Chains
- Stock, Flow and Feedback Maps
- System Dynamics models (simulation)

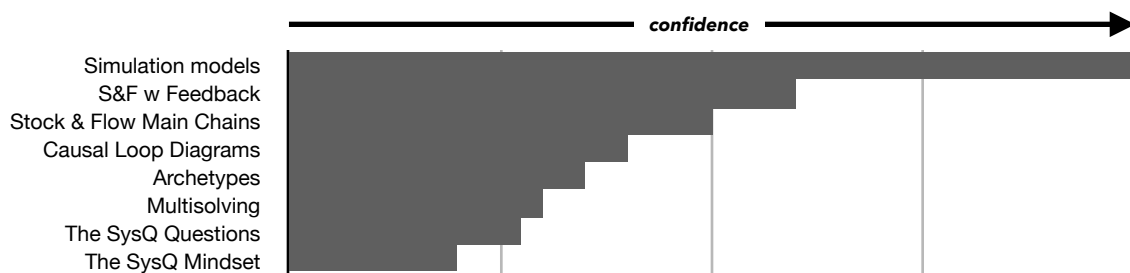
"Man must shape his tools lest they shape him."

–Arthur Miller

Many of these SysQ tools have been in the standard systems thinking kit for years; colleagues and I have more recently developed others to make it easier to effectively use SysQ. For example, the SysQ Questions were developed when I was teaching public health practitioners how to quickly improve the level of systemic rigor brought to a conversation *without having to use* some of the more time consuming visual tools.

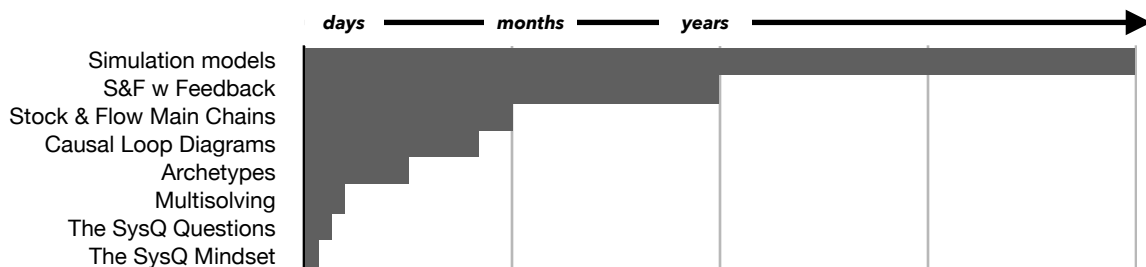
There are at least two useful ways to **assess** the **value** of these **tools**. One way relates to the **purpose** of using a tool or technique. The purpose of all the tools / techniques is to **improve** the **usefulness** (quality, rigor, accuracy) of the **mental** model **you** and your colleagues **develop during** the **SysQ process**. In other words, the tool should **improve** your **confidence** that you **understand** the **structural forces** – the essential elements of reality – generating the performance you wish to influence / improve.

Building **simulation models**, if done well, will **create** the **most confidence** – because they must be internally consistent and use quantifiable data. Developing stock and flow maps can also greatly increase confidence, because they form the basis of simulation models and could be simulatable if data were added to them. But because we are often unable to predict how the dynamics in a map would unfurl without a computer, we have less confidence in our insight than if converted to a simulation model. The chart below isn't based on data research nor contains numerical precision.



Improved Confidence in Mental Model

Another **attribute** of the range of tools / techniques is how much is **time required to learn** and **apply**. It's a basic **tenet** of SysQ that those tools that are **easy** to **learn** usually require **less time to apply** and use. For example, people can learn the SysQ Questions in a few hours and immediately apply them in meetings. The **skills** of a competent **simulation modeler** take months or **years to learn** and it can take several **months** working with an organization **to develop** a **simulation model** that generates insight with confidence.



(Relative) Time to Learn Tools and Techniques

Skilled SysQ practitioners introduce and teach a specific tool using a “just in time, just what’s needed” approach. Clients apply tools in the appropriate order once they’ve set the desired learning objectives and product outcomes.

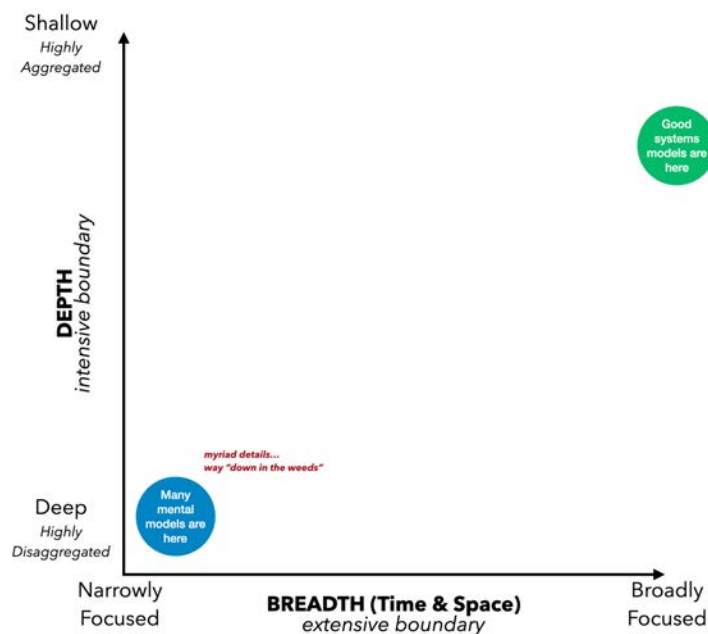
The Process Redux: The Learning Journey Takes the High Road

We **live in** the **details**. We most easily see the “here and now”. Most PhDs are granted for magnifying a weed within a field, and dissecting that weed even further.

Good systems models are at **50,000 feet**...seeing the whole. Good systems models analyze long term **patterns** and **trends** – seeking to make things better and better over the **long term**.

In the diagram here, most mental models are in the lower left corner. Good systems mental models are in the upper right: they are **broad in space and time** and **aggregated to 50,000 feet** to see the **full ecosystem**. The goal for any individual, team, coalition, or community is to

develop a good collective systems mental model – that will help identify leverage.¹¹



How Broad (Time & Space)?

Most mental models are limited in their scope of time and space. We primarily focus on days or weeks, while the **system operates over years** (in terms of time). Similarly, we concentrate on our department or division, but influencing system performance necessitates **managing multiple agencies** and **stakeholders** (in terms of space).

How Deep (Detailed)?

You may have heard adaptive leaders should rise above the dance floor and spend time on the balcony. It sometimes feels like we are below the dance floor. It's easier to see the differences between individuals in the organization than it is to see the shared worldview (culture) we all apply to how we interact with outsiders... with those we serve. We prefer a tree-by-tree approach (TREE Thinking) over the ability to look at the woods – **apply FOREST Thinking**. Public health professionals must manage large populations. Marketing departments must manage large customer segments.

“A model should be as simple as possible, but no simpler.”

–Albert Einstein

How to Build Good Systems Models

Good systems models are in the upper right. They see broadly to **understand** the forces across the **full ecosystem** responsible for the performance issues you want to improve. The goal is to generate good systems models. The challenge? **How** to get there!

There are **two roads** we could take to reach the destination of a useful systems model.

Road 1: Representing the System

Road 1, the most traveled path, tries **building** a **comprehensive map** of the entire system – **piece by piece** – from the outset. While it attempts to address crucial issues, teams composed of experts with narrow and specific knowledge often attempt to **integrate everything** they know into a **detailed picture**. However, this approach can lead to **overly complex maps**, akin to the Afghanistan Counterinsurgency map, with an excessive focus on mapping every detail. This pursuit of a complete system map places us on the **Low Road** (Road 1).

The Low Road begins with **individual ideas** and attempts to **assemble them** into a coherent whole. Given our current level of detail and the overwhelming nature of the task, this process only **exacerbates** the **complexity**. If fortunate, the arduous mapping project concludes with a map as intricate as the counterinsurgency map – a tangled web of information. More likely, the mapping **effort stalls out**, leaving us with incomplete and unhelpful insights.



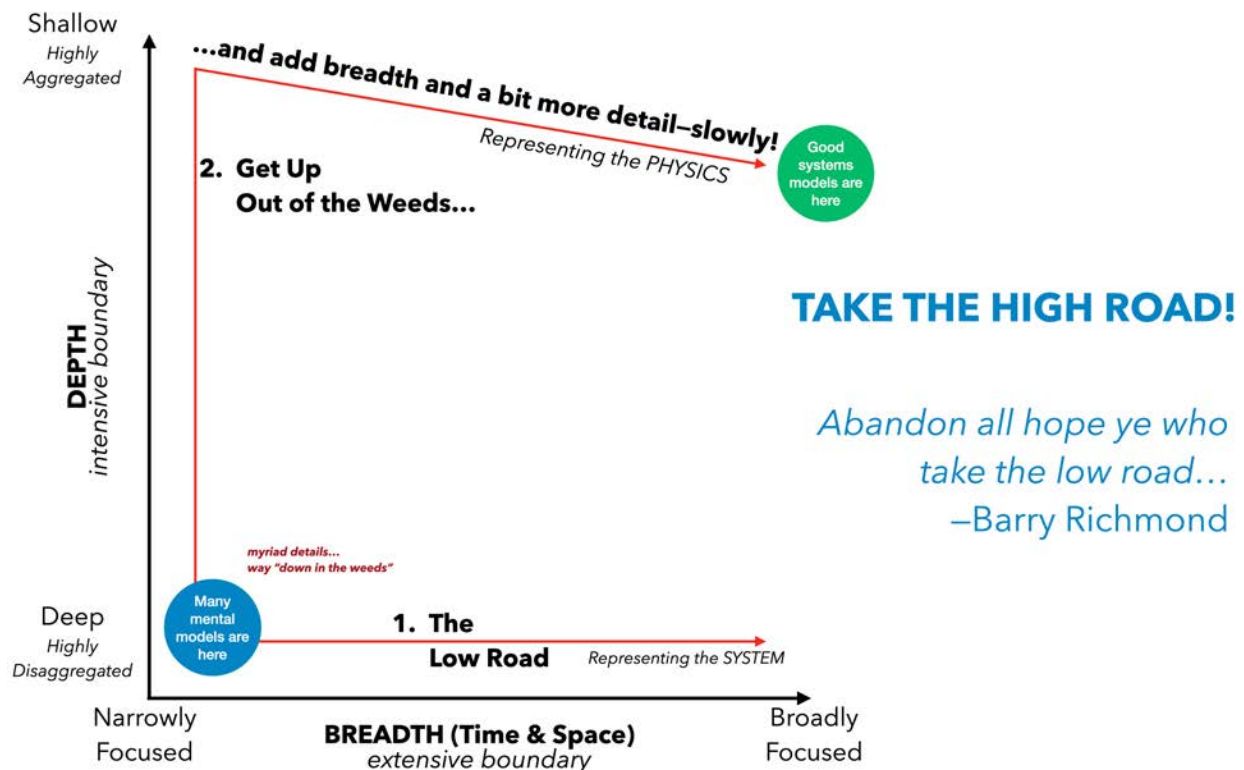
“Details are confusing. It is only by selection, by elimination, by emphasis, that we get at the real meaning of things.”

–Georgia O’Keefe

The problem stems from trying to map the system versus trying to **understand only the cause of the behavior** (performance measure) we want to improve. Because we only want to reduce the Slinky’s oscillation, we only need to **focus on** the **physics** of the spring. We can leave out colors and shapes – in fact, we must **exclude** them to reduce **complexity**. For usability, we must **eliminate parts of the full system that don’t contribute to the performance we want to improve**.

Road 2: Get Up Out of the Weeds

The wisest path – Road 2 – initially **sets aside** all the **details** and sets the **big picture vision** for system performance. In the future, what do you want to see improve? The process continues by building a **simple** causal **map**, selecting and **revising** an appropriate system **archetype**, or **creating** a **simulation model**. These **tools** help explain what forces generate that behavior.



The process continues by **sharing** the “starter map” or “starter model” **with others**. Does it explain the causes of the performance issue? What needs to be changed or added? **Make small changes. Share and test again.**

Keep adding a little bit of detail at a time. This is the downward slope in the diagram labeled “add breadth and a bit more detail – slowly”.

Stop mapping (or other SysQ analysis process) when your answer to the question “Does this **describe** the essential **drivers of** the **performance** issue?” is “Yes!” At that point you begin to **look for leverage points**

“Two roads diverged in a wood, and
I – I took the one less traveled by,
And that has made all the
difference”

—Robert Frost

SysQ Facilitator Role

In working with a SysQ practitioner, your **tendency** will be to **increase** the **complexity** of the maps, models, and analyses collaboratively generated. You will prefer a **“devil’s in the details” mindset** – TREE Thinking over FOREST Thinking. The practitioner’s job is to help you apply Systemic Intelligence to your analysis and strategy: the **SysQ process** is designed to **reduce** the **complexity** and **noise** in ways that helps you **generate actionable understanding** to solve complex adaptive challenges.

There is **tension between typical client biases** they bring to developing strategy **and** the **SysQ objectives** guiding the practitioner’s work with them. This table describes typical biases, the SysQ objective, and how that tension will often feel to the client.

Client Bias	SysQ Objective	How it will feel
<i>We must increase the detail complexity of the maps, models, and analyses we collaborate on.</i>	The process objective is to reduce detail complexity and noise in ways that help you generate actionable understanding to solve complex adaptive challenges	<i>This is too simple! Our reality is much more complex.</i>
<i>We want you to build a product for us – with little involvement of effort, resources and time from us.</i>	The process must build your capacity along the way. This helps collaboratively create a better product – and you’re able to use and explain insights after the project ends.	<i>This is hard work – can’t you give us an answer?</i>
<i>Make sure this process affirms our current or recently developed strategy.</i>	The practitioner’s role is to rigorously challenge the assumptions made during the process of strategy-building – to increase confidence your final strategy works “as is” or determine if it needs revision.	<i>We can’t question the strategy – we’re already implementing it. Can’t you just help improve it slightly?</i>
<i>We can’t start now. We must wait until all data is gathered before beginning a SysQ process.</i>	Rigorous application of the SysQ process will identify data needs – including data gaps – to move into strategy development sooner. Further, this will create a more streamlined data agenda.	<i>We’re not ready to start yet. We need more time for a landscape analysis or more research or...</i>

Client Bias	SysQ Objective	How it will feel
<i>Let's start with generating solutions.</i>	The SysQ process facilitates a "go slow to go fast" process. We begin by ensuring the problem is suitably defined for SysQ. Then we advance to building an understanding of the problem – <i>before</i> attempting to generate solutions.	<i>This is taking too long! We urgently need to find solutions.</i>
<i>The consultant should integrate and interpret assumptions about the problem or ecosystem to improve exactly as client prefers.</i>	Because we often apply low SysQ to building mental models, the SysQ practitioner will apply the SysQ thinking skills to question client assumptions about their strategic goals and understanding of the issues. The goal: transform client mental models to conform to SysQ analysis	<i>That's not the way to map or model the problem.</i>

""Faced with the choice between changing one's mind and proving that there is no need to do so, almost everyone gets busy on the proof."

–John Kenneth Galbraith

Conclusion

Systemic Intelligence offers a powerful approach to **understanding** and **addressing complex challenges** in our increasingly interconnected world. By focusing on the big picture and identifying the essential **drivers of performance**, it enables individuals and organizations to move beyond surface-level solutions and create **lasting**, positive **change**. The process of developing good systems mental models – though requiring patience and a shift in perspective – ultimately leads to more **effective problem-solving** and decision-making.

Learning and applying Systemic Intelligence can benefit professionals across various fields, from **business leaders** and **policymakers** to **educators** and **healthcare professionals**. By adopting this approach, you can enhance your ability to navigate complexity, anticipate long-term consequences, and identify **high-leverage interventions**. Moreover, the skills developed through practicing Systemic Intelligence – such as holistic thinking, pattern recognition, and strategic analysis – are increasingly valuable in today's rapidly evolving professional landscape.

As we face unprecedented global challenges, from climate change to economic inequality, **the need for systemic thinking has never been greater**. By building our Systemic Intelligence, we can cultivate a more rigorous understanding of the world around us and develop innovative solutions that **address root causes** rather than symptoms. Whether you're tackling organizational issues, societal problems, or personal challenges, the principles of Systemic Intelligence offer a roadmap for more effective, **sustainable**, and impactful **decision-making**.

About the Author

Chris Soderquist helps people and teams approach their most pressing challenges with more focus and force. He has developed the field of *Systemic Intelligence* – SysQ – the capacity to apply tools and concepts of systems thinking to design high-leverage interventions for addressing real-world problems. His unique work elevates traditional "systems thinking" into a discipline that is easier to learn, easier to apply, and more impactful. This discipline helps clients build high performing teams and projects that deliver on their objectives in a more focused, efficient, and effective way.

To learn more visit findinghighleverage.com.



Endnotes

¹ Richmond, B. (2000). *The "thinking" in systems thinking: Seven essential skills*. Pegasus Communications.

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⁴ Heifetz, R. *Leadership without Easy Answers*. Cambridge, Mass. :Belknap Press of Harvard University Press, 1994.

⁵ Tenor Gif, <https://tenor.com/view/pro-speed-eating-fast-loser-wtf-gif-16379929>

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¹⁰ Clark, K. et al. (July 2015). *Using Systems Thinking to Shift Mindsets*. [white paper]. International System Dynamics Conference, Boston, MA, USA.

¹¹ Diagram first developed by Barry Richmond, *isee systems*.

